

MSE 311 Thermodynamics of materials-Homework 7

Name:

Student ID:

1. An ideal Diesel cycle has a compression ratio of 17 and a cutoff ratio of 1.3. Determine the maximum temperature of the air and the rate of heat addition to this cycle when it produces 140 kW of power and the state of the air at the beginning of the compression is 90 kPa and 578°C. Use constant specific heats at room temperature ($c_p=1$ kJ/kg/K, $c_v=0.718$ kJ/kg/K).

$$r = \frac{V_1}{V_2} = 17, r_c = \frac{V_3}{V_2} = 1.3, k = \frac{c_p}{c_v} = \frac{1}{0.718} = 1.39$$

$$w = q_{in} - q_{out} = 140 \text{ kW}, P_1 = 90 \text{ kPa}, T_1 = 578^\circ\text{C} = 851 \text{ K}$$

$$T_3 = \frac{P_3 V_3}{nR} = \frac{RT_1}{P_1 V_1} \frac{P_2 V_3}{R} = T_1 \frac{P_2}{P_1} \frac{V_3}{V_2} \frac{V_2}{V_1} = T_1 \left(\frac{V_1}{V_2} \right)^{k-1} \frac{V_3}{V_2} = T_1 (r)^{k-1} r_c = 851 \times 17^{1.39-1} \times 1.3 = 3340 \text{ K}$$

$$\eta = 1 - \frac{1}{r^{k-1}} \left[\frac{r_c^k - 1}{k(r_c - 1)} \right] = 1 - \frac{1}{17^{1.39-1}} \times \left[\frac{1.3^{1.39} - 1}{1.39 \times (1.3 - 1)} \right] = 0.65$$

$$q_{in} = \frac{w}{\eta} = \frac{140}{0.65} = 215 \text{ kW}$$

2. A four-cylinder, four-stroke, 2.2-L gasoline engine operates on the Otto cycle with a compression ratio of 10. The air is at 100 kPa and 60°C at the beginning of the compression process, and the maximum pressure in the cycle is 8 MPa. The compression and expansion processes may be modeled as polytropic with a polytropic constant of 1.3. Using constant specific heats at 850 K ($c_p = 1.11$ kJ/kg/K, $c_v = 0.823$ kJ/kg/K), determine (a) the temperature at the end of the expansion process, (b) the net work output and the thermal efficiency, (c) the engine speed for a net power output of 70 kW, and (d) the specific fuel consumption, in g/kWh, defined as the ratio of the mass of the fuel consumed to the net work produced. The air-fuel ratio, defined as the amount of air divided by the amount of fuel intake, is 16.

$$r = \frac{V_1}{V_2} = 10, k = 1.3$$

$$P_1 = 100 \text{ kPa}, T_1 = 60^\circ\text{C} = 333 \text{ K}, P_3 = 8000 \text{ kPa}$$

$$(a) \quad T_4 = \frac{P_4 V_4}{nR} = \frac{RT_1}{P_1 V_1} \frac{P_4 V_4}{R} = \frac{T_1}{P_1} P_4 = \frac{T_1}{P_1} P_3 \left(\frac{V_3}{V_4} \right)^k = \frac{T_1}{P_1} P_3 r^{-k} = \frac{333}{100} \times 8000 \times 10^{-1.3} = 1335 \text{ K}$$

$$\eta = 1 - \frac{1}{r^{k-1}} = 1 - \frac{1}{10^{1.3-1}} = 0.499$$

$$(b) \quad q_{out} = c_v (T_4 - T_1) = 0.823 \times (1335 - 333) = 825 \text{ kJ / kg}$$

$$\eta = \frac{w}{q_{in}} = \frac{w}{w + q_{out}} \Rightarrow w = \frac{\eta q_{out}}{1 - \eta} = \frac{0.499 \times 825}{1 - 0.499} = 822 \text{ kJ / kg}$$

$$r = \frac{V_r + V_s}{V_r} \Rightarrow V_r = \frac{V_s}{r-1} = \frac{2.2}{10-1} = 0.244L$$

$$V_1 = V_r + V_s = 0.244 + 2.2 = 2.444L$$

$$(c) \quad m = nM = \frac{P_1 V_1 M}{MRT_1} = \frac{100 \times 10^3 \times 2.444 \times 10^{-3} \times 28.96}{8.314 \times 333} = 2.556g$$

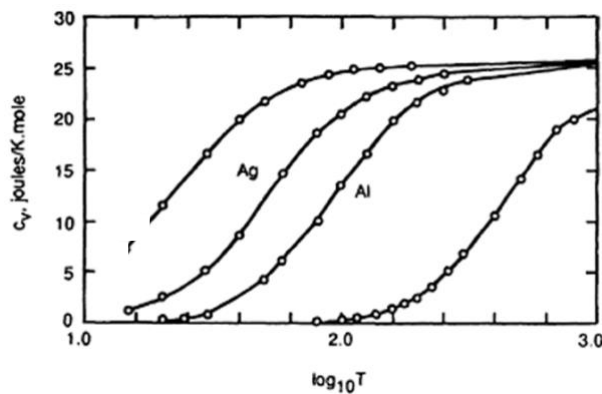
$$n = 2 \frac{P}{wm4} = \frac{70}{822 \times 2.556 \times 10^{-3} \times 2} = 16.66r / s = 1000r / \min$$

$$\frac{m - m_f}{m_f} = 16 \Rightarrow m_f = \frac{m}{17} = \frac{2.556}{17} = 0.15g$$

$$(d) \quad sfc = \frac{m_f}{mw} = \frac{0.15}{2.556 \times 10^{-3} \times 822} = 0.0714g / kJ = 257g / kWh$$

3. It is well-known that the heat capacity follows the Dulong-Petit law, but it significantly deviates from the value of $3R$. Figure below shows the temperature dependent heat capacity, c_v , of four solid elements.

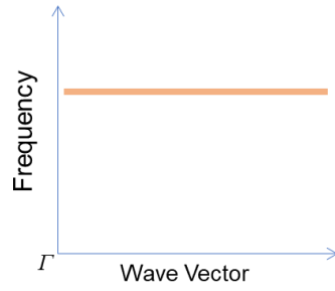
- (1) Please write the approximate relationship of c_v and T , as T is close to $0K$.
- (2) Please tell the differences between Einstein model and Debye model of c_v , you can explain with schematic figure. Which model is better at low temperatures?
- (3) Use the relation to distinguish and mark the two curves between Ag ($\theta_D = 213$) and Al ($\theta_D = 385$) in the image below.



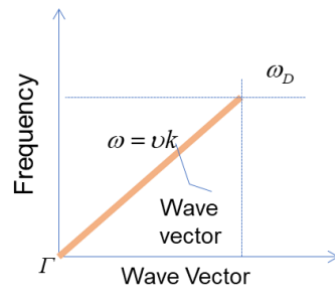
$$(1) \quad c_v = 9R \left(\frac{T}{\theta_D} \right)^3 \int_0^{\infty} \frac{x^4 e^x}{(e^x - 1)^2} dx$$

(2) Einstein's assumption: the assumption the oscillators vibrate with a single frequency

Debye's modification: the oscillators vibrate with in a rang of frequency



Einstein model:



Debye model:

Debye model is better at low temperature